VLM Finetuning Experiment

Vision-Language Model Fine-Tuning Experiment

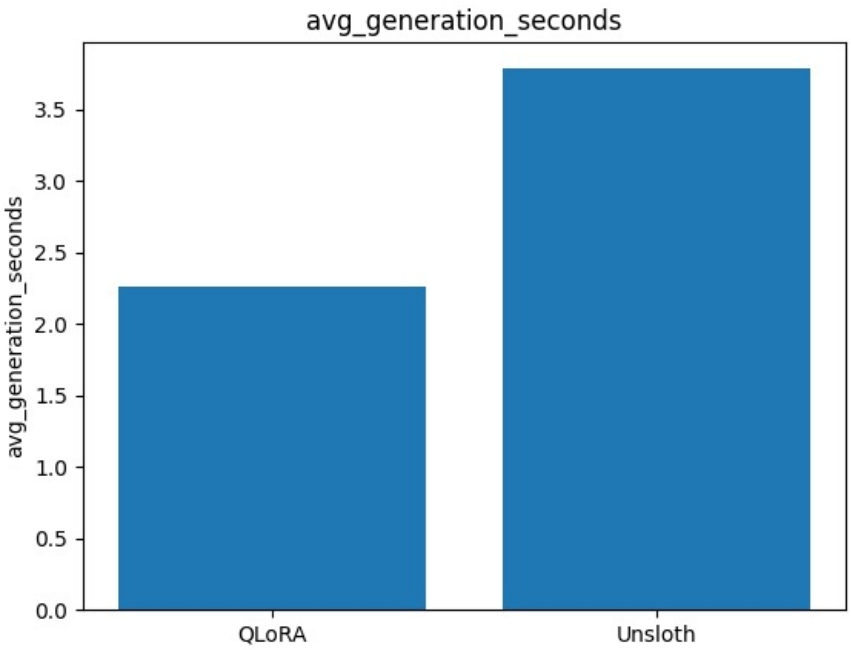
Experiment Setup

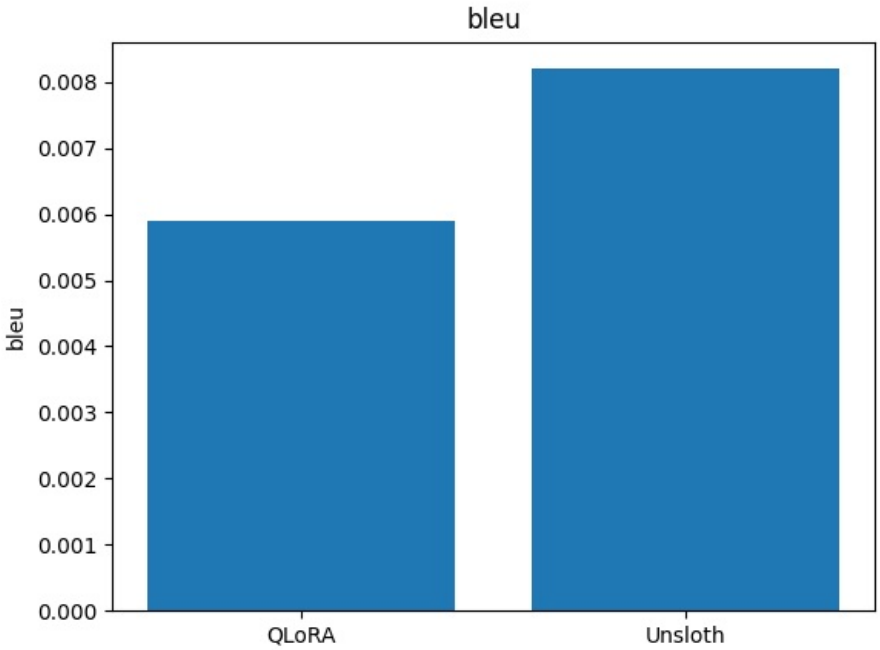
- Current cleaned dataset size: 65 samples
- Model: LLaVA 1.5 7B
- Training methods compared: QLoRA vs Unsloth

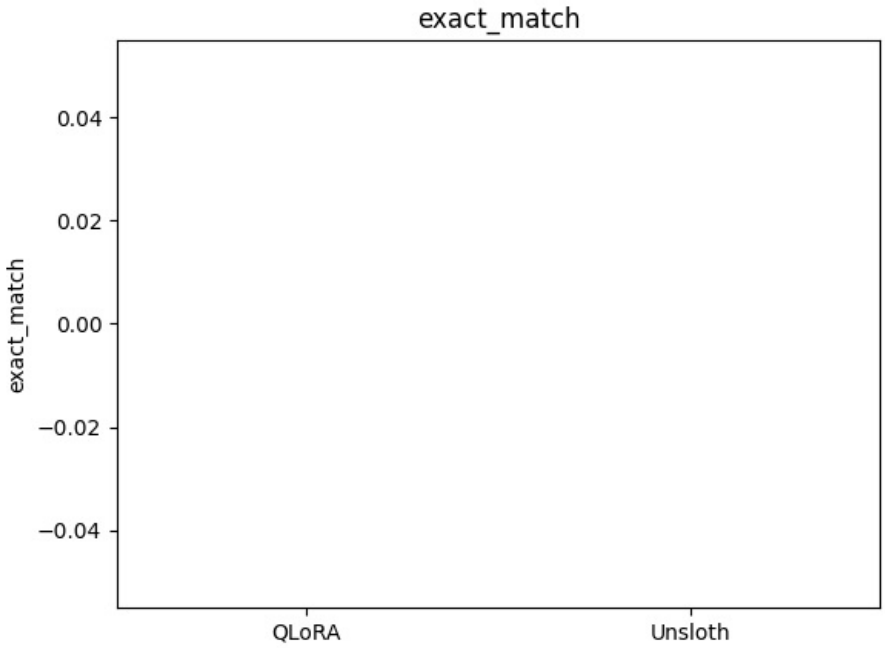
Aggregate Results

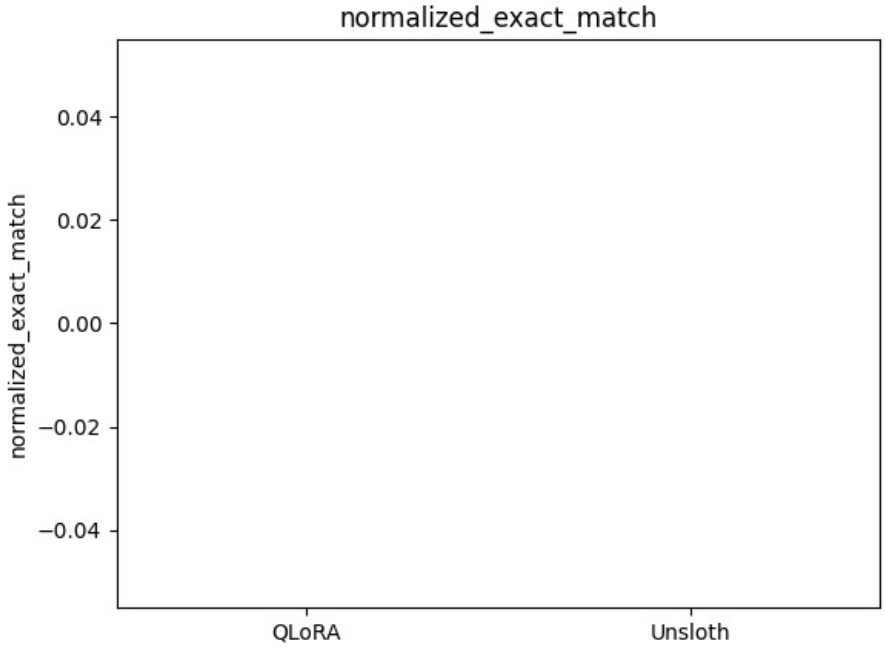
Metric	QLoRA	Unsloth
avg_generation_seconds	2.2602	3.7867
bleu	0.0059	0.0082
exact_match	0.0	0.0
normalized_exact_match	0.0	0.0
peak_vram_gb	12.41	4.68
rouge1	0.1152	0.1186
rouge2	0.0154	0.0191
rougeL	0.0963	0.1024
samples_evaluated	65	65
skipped_invalid_images	0	0
tokens_per_second	1245.66	452.72
total_generation_seconds	146.91	246.13
training_time_seconds	30.74	84.59

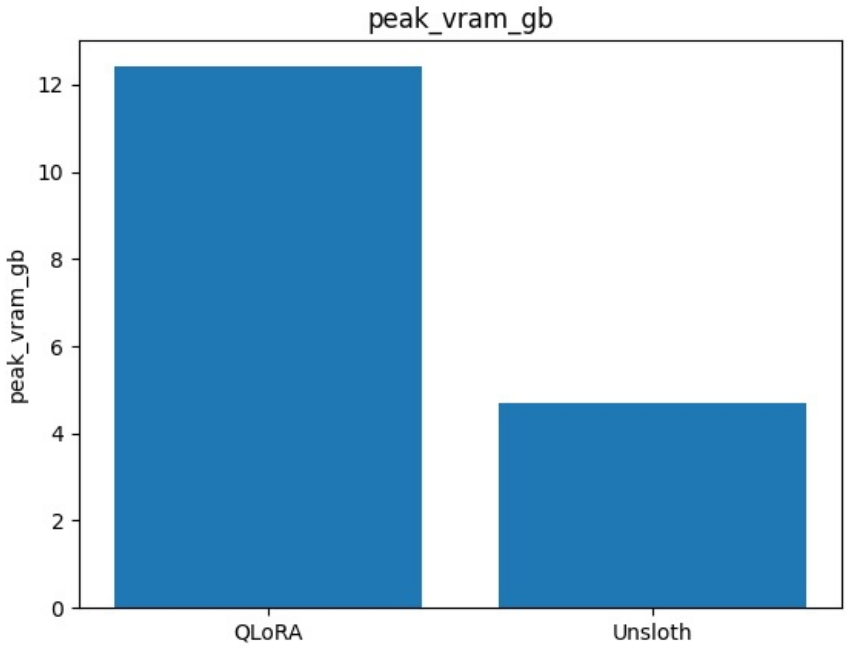
Training And Evaluation Charts

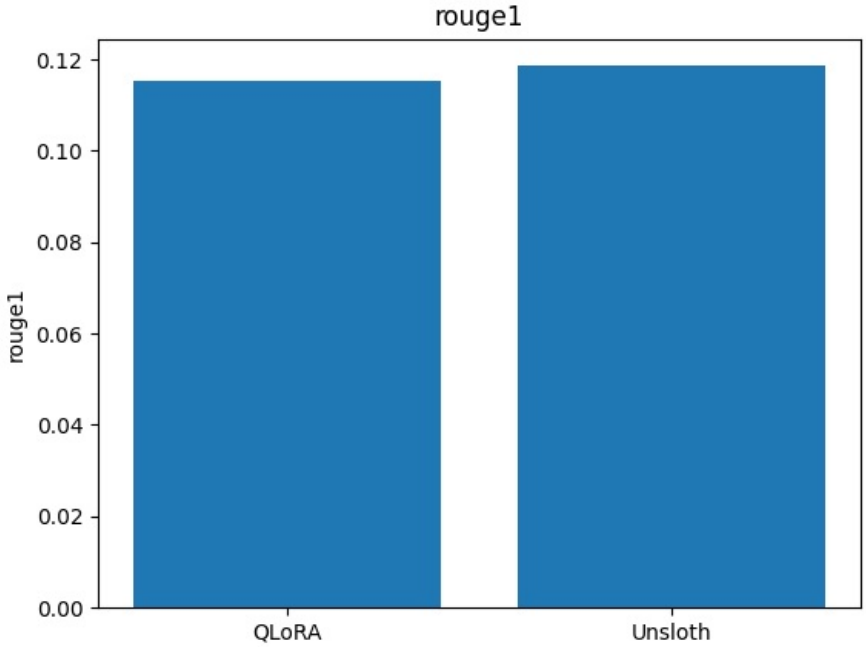

avg_generation_seconds
avg_generation_seconds
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
QLoRA
Unsloth

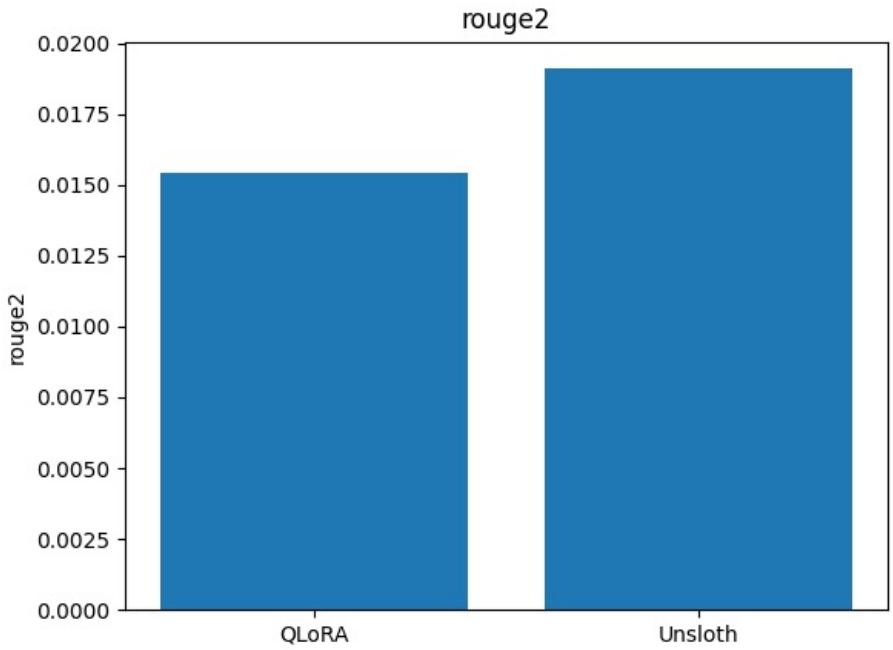

bleu
bleu
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0.000
QLoRA
Unsloth

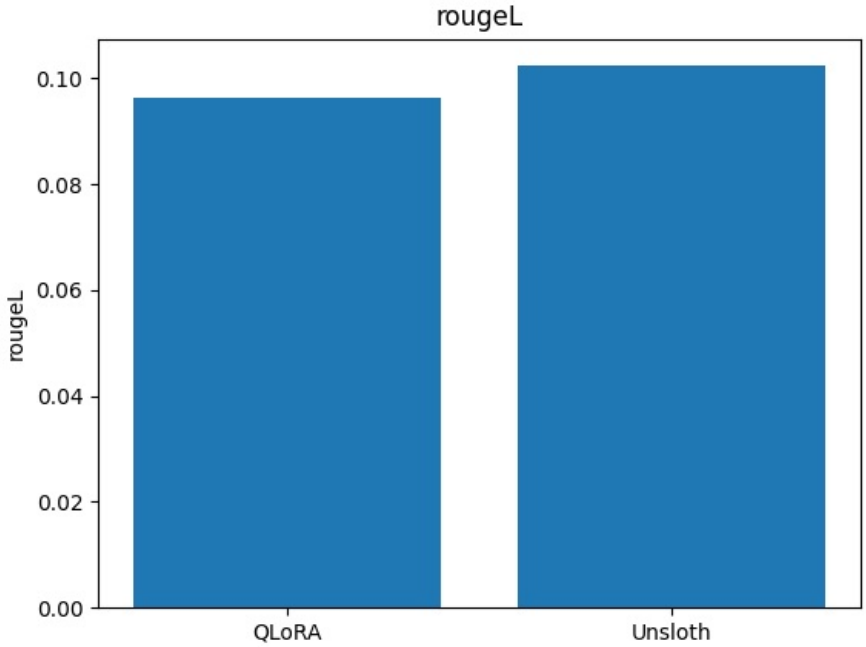

exact_match
exact_match
0.04
0.02
0.00
−0.02
−0.04
QLoRA
Unsloth

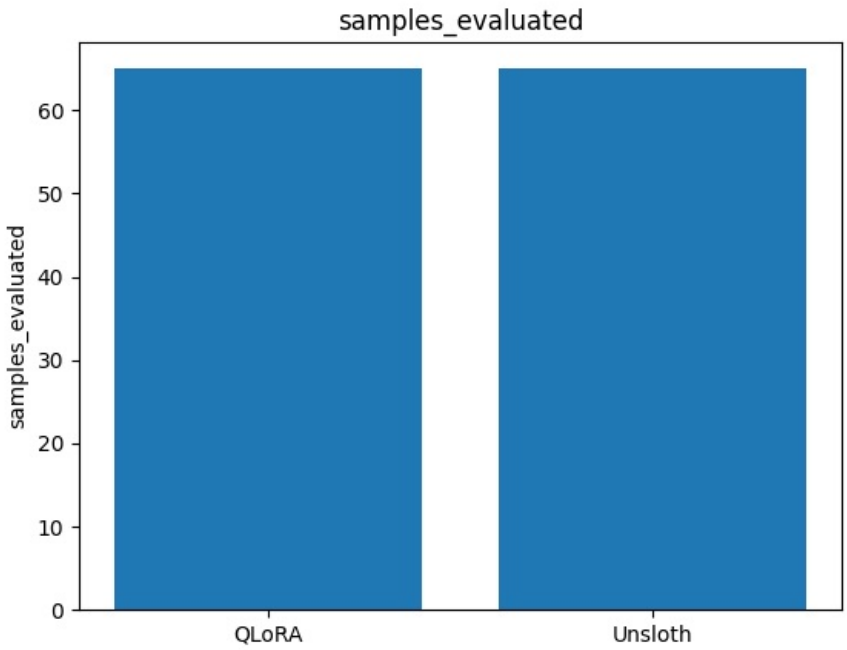

normalized_exact_match
normalized_exact_match
0.04
0.02
0.00
−0.02
−0.04
QLoRA
Unsloth

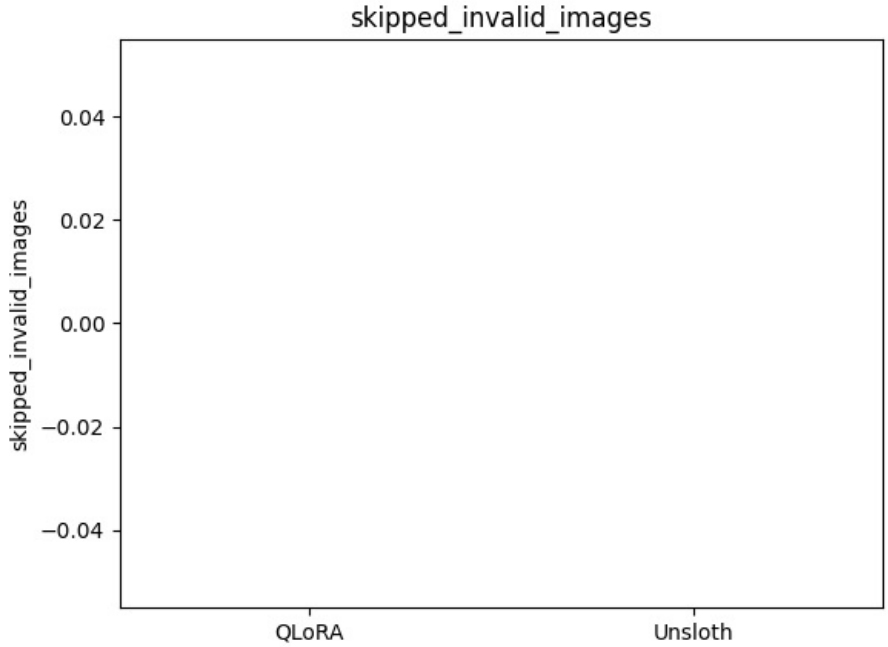

peak_vram_gb
peak_vram_gb
0
2
4
6
8
10
12
QLoRA
Unsloth

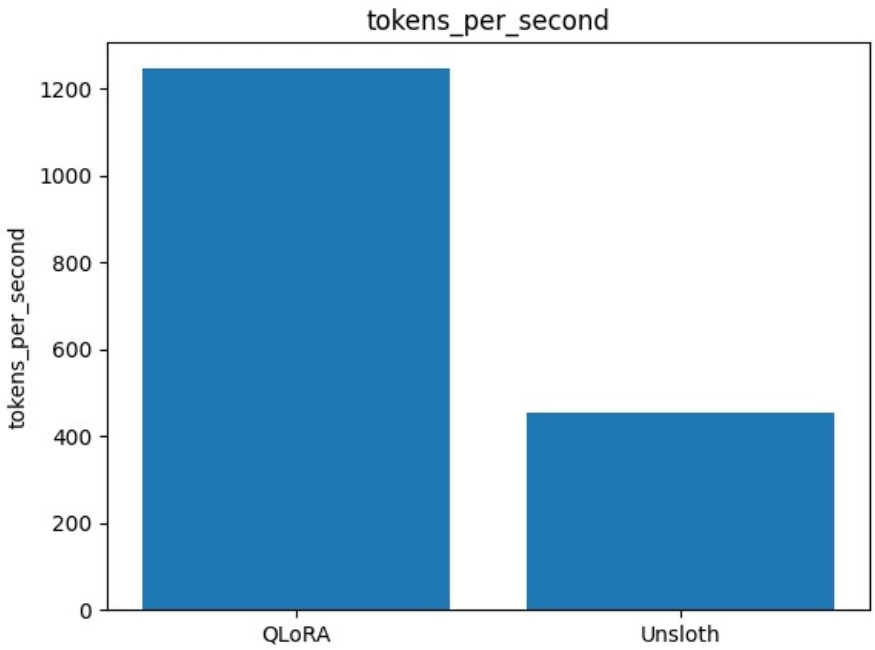

rouge1
rouge1
0.00
0.02
0.04
0.06
0.08
0.10
0.12
QLoRA
Unsloth

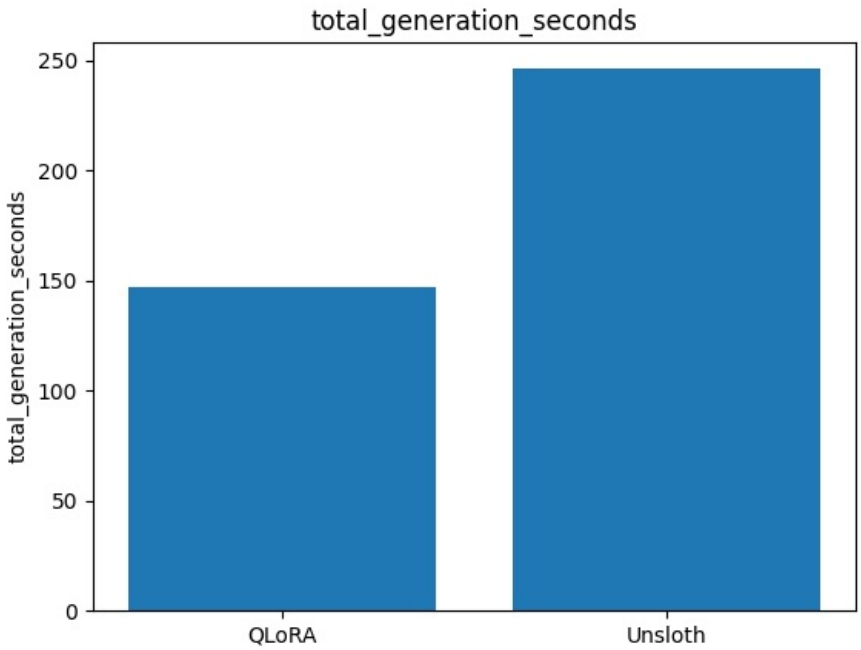

rouge2
0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
rouge2
QLoRA
Unsloth

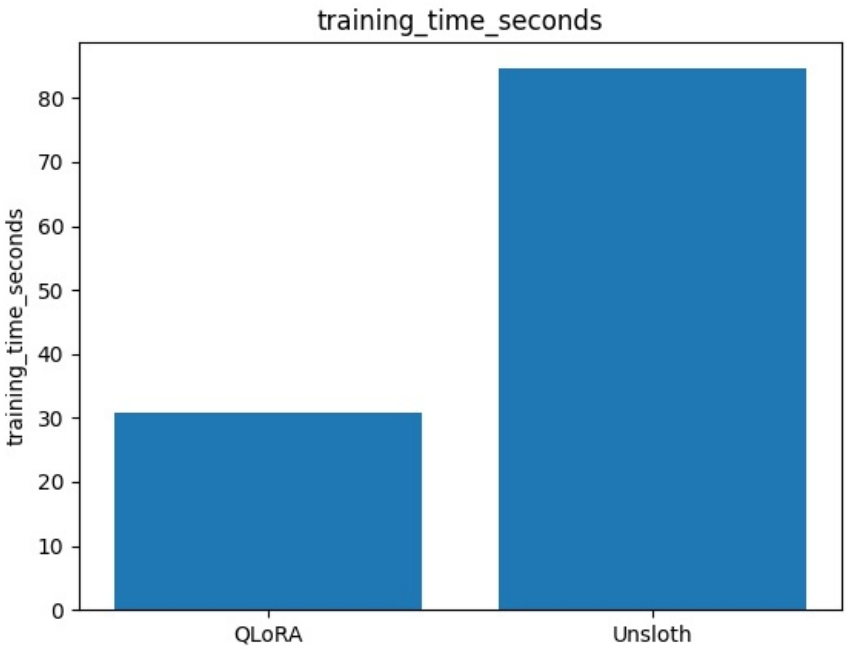

rougeL
0.10
0.08
0.06
0.04
0.02
0.00
rougeL
QLoRA
Unsloth


samples_evaluated
samples_evaluated
0
10
20
30
40
50
60
QLoRA
Unsloth


skipped_invalid_images
skipped_invalid_images
0.04
0.02
0.00
−0.02
−0.04
QLoRA
Unsloth


tokens_per_second
tokens_per_second
1200
1000
800
600
400
200
0
QLoRA
Unsloth


total_generation_seconds
total_generation_seconds
250
200
150
100
50
0
QLoRA
Unsloth



Example Predictions

- QLoRA prediction: The image depicts a bustling bus station with numerous buses parked in a parking lot. The buses are of various sizes and are scattered throughout the scene. There are also several people walking around the station, likely
- Unsloth prediction: The image depicts a bustling bus station with numerous buses and people. There are at least 13 buses of various sizes and colors, parked in a parking lot or on the street. The buses

Observations

The benchmark reports both efficiency metrics and dataset-wide caption quality metrics on the current cleaned dataset.